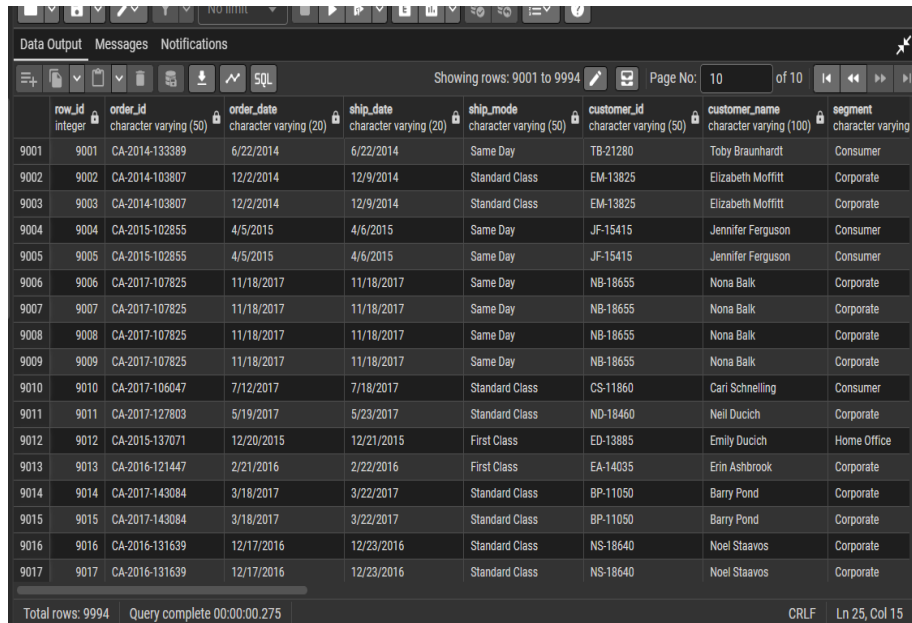


BRIEF INSIGHTS + QUERY RESULTS OF SQL SCRIPT

1. **Load Dataset into SQL Database** - This step creates the table structure and prepares the database for storing sales records.

Output-

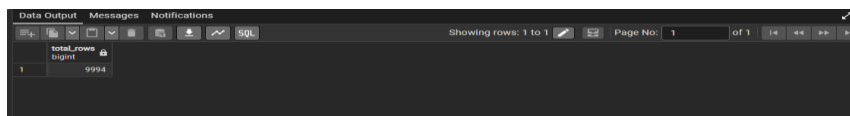


row_id	order_id	order_date	ship_date	ship_mode	customer_id	customer_name	segment
9001	CA-2014-133389	6/22/2014	6/22/2014	Same Day	TB-21280	Toby Braunhardt	Consumer
9002	CA-2014-103807	12/2/2014	12/9/2014	Standard Class	EM-13825	Elizabeth Moffitt	Corporate
9003	CA-2014-103807	12/2/2014	12/9/2014	Standard Class	EM-13825	Elizabeth Moffitt	Corporate
9004	CA-2015-102855	4/5/2015	4/6/2015	Same Day	JF-15415	Jennifer Ferguson	Consumer
9005	CA-2015-102855	4/5/2015	4/6/2015	Same Day	JF-15415	Jennifer Ferguson	Consumer
9006	CA-2017-107825	11/18/2017	11/18/2017	Same Day	NB-18655	Nona Balk	Corporate
9007	CA-2017-107825	11/18/2017	11/18/2017	Same Day	NB-18655	Nona Balk	Corporate
9008	CA-2017-107825	11/18/2017	11/18/2017	Same Day	NB-18655	Nona Balk	Corporate
9009	CA-2017-107825	11/18/2017	11/18/2017	Same Day	NB-18655	Nona Balk	Corporate
9010	CA-2017-106047	7/12/2017	7/18/2017	Standard Class	CS-11860	Cari Schnelling	Consumer
9011	CA-2017-127803	5/19/2017	5/23/2017	Standard Class	ND-18460	Neil Duchich	Corporate
9012	CA-2015-137071	12/20/2015	12/21/2015	First Class	ED-13885	Emily Duchich	Home Office
9013	CA-2016-121447	2/21/2016	2/22/2016	First Class	EA-14035	Erin Ashbrook	Corporate
9014	CA-2017-143084	3/18/2017	3/22/2017	Standard Class	BP-11050	Barry Pond	Corporate
9015	CA-2017-143084	3/18/2017	3/22/2017	Standard Class	BP-11050	Barry Pond	Corporate
9016	CA-2016-131639	12/17/2016	12/23/2016	Standard Class	NS-18640	Noel Staavos	Corporate
9017	CA-2016-131639	12/17/2016	12/23/2016	Standard Class	NS-18640	Noel Staavos	Corporate

2. **Explore table-** In this particular step we explore table like Total ROWS in table, display Sample Data (eg- LIMIT 10) as well as Schema / Structure.

1. **Total Rows in Table**-This query counts the total number of records imported into the table.

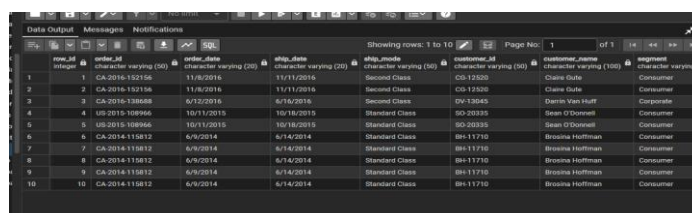
Output-



total_rows
9994

2. **Sample Data (LIMIT 10)**-This query displays the first 10 records for quick data inspection.

Output-



row_id	order_id	order_date	ship_date	ship_mode	customer_id	customer_name	segment
1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CS-12320	Chloe Gulle	Consumer
2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CS-12320	Chloe Gulle	Consumer
3	CA-2016-138686	6/12/2016	6/16/2016	Standard Class	DN-13045	Danish Van Hout	Corporate
4	US-2010-109966	10/11/2015	10/18/2015	Standard Class	SD-20335	Sean O'Donnell	Consumer
5	US-2015-109966	10/11/2015	10/18/2015	Standard Class	SD-20335	Sean O'Donnell	Consumer
6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffmann	Consumer
7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffmann	Consumer
8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffmann	Consumer
9	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffmann	Consumer
10	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffmann	Consumer

3. **Schema / Structure**- This query shows column names and their data types in the table

Output-

	column_name	data_type
	name	character varying
1	row_id	integer
2	order_id	character varying
3	order_date	character varying
4	ship_date	character varying
5	ship_mode	character varying
6	customer_id	character varying
7	customer_name	character varying
8	segment	character varying
9	country	character varying
10	city	character varying
11	state	character varying
12	postal_code	character varying
13	region	character varying
14	product_id	character varying
15	category	character varying
16	sub_category	character varying
17	product_name	character varying
18	price	numeric

Total rows: 21 Query complete 00:00:00.206 CRLF Ln 37, Col 25

3. **Apply WHERE** – Where is used basically for filtering the data.

1. **Filter by Region (West)**- This query retrieves all sales records belonging to the West region.

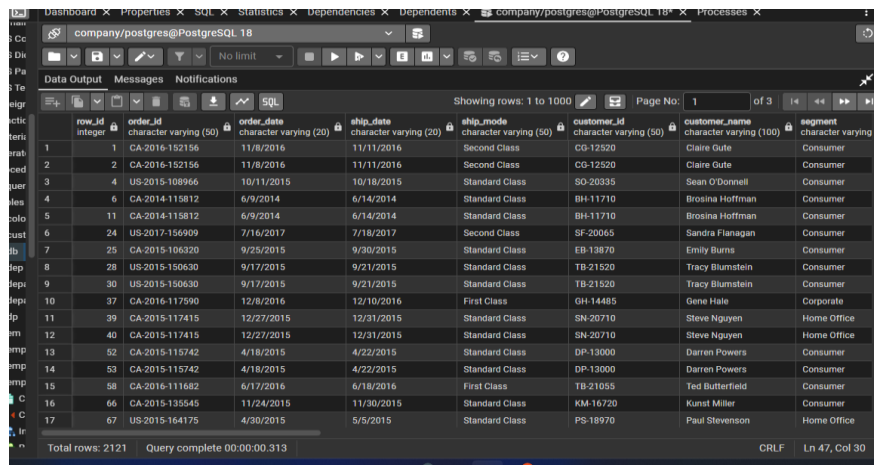
Output-

row_id	order_id	order_date	ship_date	ship_mode	customer_id	customer_name	segment	country	city
1	3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	
2	6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	
3	7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	
4	8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	
5	9	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	
6	10	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	
7	11	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	
8	12	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	
9	14	CA-2016-161389	12/5/2016	12/10/2016	Standard Class	IM-15070	Irene Maddox	Consumer	
10	18	CA-2014-167164	5/13/2014	5/15/2014	Second Class	AG-10270	Alejandro Grove	Consumer	
11	19	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	
12	20	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	
13	21	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	
14	25	CA-2015-106320	9/25/2015	9/30/2015	Standard Class	EB-13870	Emily Burns	Consumer	
15	26	CA-2016-121755	1/16/2016	1/20/2016	Second Class	EH-13945	Eric Hoffmann	Consumer	
16	27	CA-2016-121755	1/16/2016	1/20/2016	Second Class	EH-13945	Eric Hoffmann	Consumer	
17	43	CA-2016-101343	7/17/2016	7/22/2016	Standard Class	RA-19885	Ruben Ausman	Corporate	

Total rows: 3203 Query complete 00:00:00.328 CRLF Ln 42, Col 23

2. **Filter by Category (Furniture)**- This query displays only records from the Furniture category.

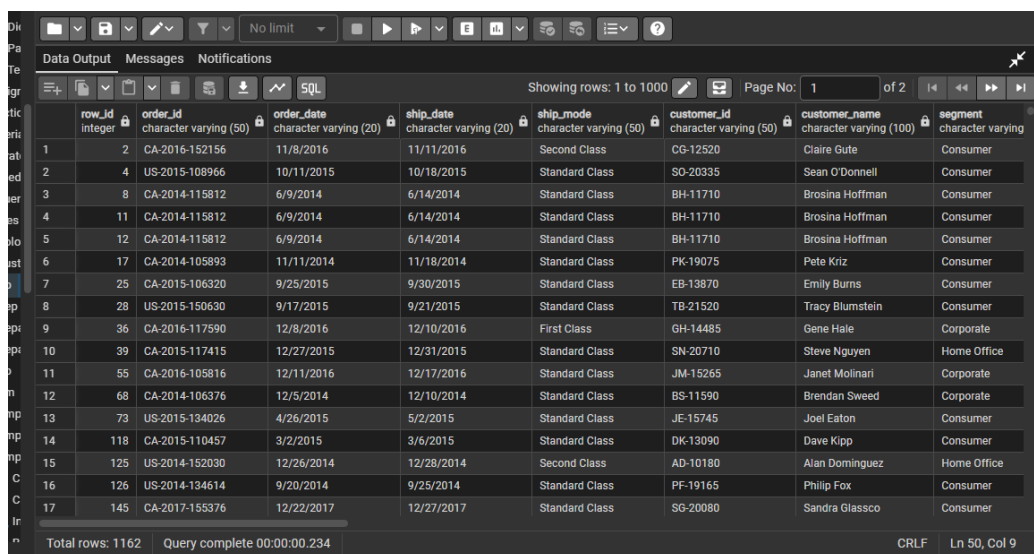
Output-



row_id	order_id	order_date	ship_date	ship_mode	customer_id	customer_name	segment
1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer
2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer
3	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer
4	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brodina Hoffman	Consumer
5	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brodina Hoffman	Consumer
6	US-2017-156909	7/16/2017	7/18/2017	Second Class	SF-20065	Sandra Flanagan	Consumer
7	CA-2015-106320	9/25/2015	9/30/2015	Standard Class	EB-13870	Emily Burns	Consumer
8	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer
9	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer
10	CA-2016-117590	12/8/2016	12/10/2016	First Class	GH-14485	Gene Hale	Corporate
11	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710	Steve Nguyen	Home Office
12	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710	Steve Nguyen	Home Office
13	CA-2015-115742	4/18/2015	4/22/2015	Standard Class	DP-13000	Darren Powers	Consumer
14	CA-2015-115742	4/18/2015	4/22/2015	Standard Class	DP-13000	Darren Powers	Consumer
15	CA-2016-111682	6/17/2016	6/18/2016	First Class	TB-21065	Isid Butterfield	Consumer
16	CA-2015-155545	11/24/2015	11/30/2015	Standard Class	KM-16720	Kunst Miller	Consumer
17	US-2015-164175	4/30/2015	5/5/2015	Standard Class	PS-18970	Paul Stevenson	Home Office

3. **Filter by Sales > 500-** This query identifies transactions with sales greater than 500.

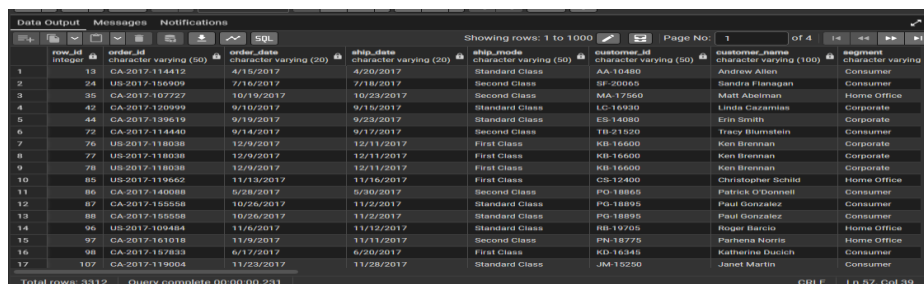
Output-



row_id	order_id	order_date	ship_date	ship_mode	customer_id	customer_name	segment
1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer
2	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer
3	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brodina Hoffman	Consumer
4	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brodina Hoffman	Consumer
5	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brodina Hoffman	Consumer
6	CA-2014-105893	11/11/2014	11/18/2014	Standard Class	PK-19075	Pete Kriz	Consumer
7	CA-2015-106320	9/25/2015	9/30/2015	Standard Class	EB-13870	Emily Burns	Consumer
8	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer
9	CA-2016-117590	12/8/2016	12/10/2016	First Class	GH-14485	Gene Hale	Corporate
10	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710	Steve Nguyen	Home Office
11	CA-2016-105816	12/11/2016	12/17/2016	Standard Class	JM-15265	Janet Molinar	Corporate
12	CA-2014-106376	12/5/2014	12/10/2014	Standard Class	BS-11590	Brendan Sweed	Corporate
13	US-2015-134026	4/26/2015	5/2/2015	Standard Class	JE-15745	Joel Eaton	Consumer
14	CA-2015-110457	3/2/2015	3/6/2015	Standard Class	DK-13090	Dave Kipp	Consumer
15	US-2014-152030	12/26/2014	12/28/2014	Standard Class	AD-10180	Alan Dominguez	Home Office
16	US-2014-134614	9/20/2014	9/25/2014	Standard Class	PF-19165	Philip Fox	Consumer
17	CA-2017-155376	12/22/2017	12/27/2017	Standard Class	SG-20080	Sandra Glassco	Consumer

4. **Filter by Date (Year 2017)** - This query extracts all orders placed during the year 2017.

Output-

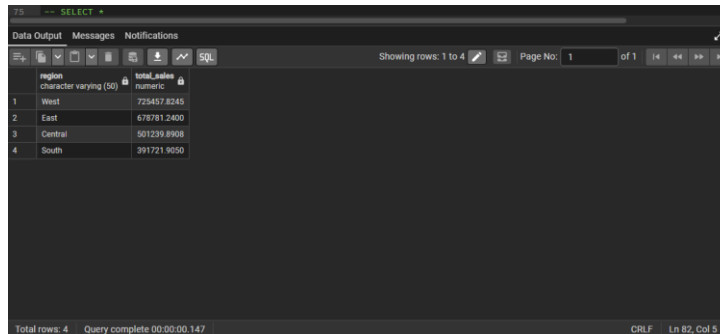


row_id	order_id	order_date	ship_date	ship_mode	customer_id	customer_name	segment
1	US-2017-156909	7/16/2017	7/18/2017	Second Class	SF-20065	Sandra Flanagan	Consumer
2	CA-2017-107727	10/19/2017	10/23/2017	Second Class	MA-17560	Matt Abelman	Home Office
3	CA-2017-120999	9/10/2017	9/15/2017	Standard Class	LC-16930	Linda Cazamas	Corporate
4	CA-2017-114440	9/19/2017	9/25/2017	Standard Class	ES-14680	Erin Smith	Corporate
5	CA-2017-118038	12/9/2017	12/11/2017	First Class	TB-21520	Tracy Blumstein	Consumer
6	US-2017-118038	12/9/2017	12/11/2017	First Class	KB-16600	Ken Brennan	Corporate
7	US-2017-118038	12/9/2017	12/11/2017	First Class	KB-16600	Ken Brennan	Corporate
8	US-2017-118038	12/9/2017	12/11/2017	First Class	KB-16600	Ken Brennan	Corporate
9	US-2017-118038	12/9/2017	12/11/2017	First Class	KB-16600	Ken Brennan	Corporate
10	US-2017-118038	12/9/2017	12/11/2017	First Class	KB-16600	Ken Brennan	Corporate
11	CA-2017-140088	5/28/2017	5/30/2017	Second Class	PO-18865	Patrick O'Donnell	Consumer
12	CA-2017-155558	10/26/2017	11/2/2017	Standard Class	PG-18895	Paul Gonzalez	Consumer
13	CA-2017-135658	10/26/2017	11/2/2017	Standard Class	PG-18895	Paul Gonzalez	Consumer
14	US-2017-109484	11/6/2017	11/12/2017	Standard Class	RB-19705	Roger Barcio	Home Office
15	CA-2017-161018	11/9/2017	11/11/2017	Second Class	PN-18775	Parthena Norris	Home Office
16	CA-2017-157833	6/17/2017	6/20/2017	First Class	KD-16345	Katherine Dulich	Consumer
17	CA-2017-119004	11/23/2017	11/28/2017	Standard Class	JM-15260	Janet Martin	Consumer

4. Use GROUP BY for aggregations-

1. **Total Sales by Region-** This query calculates total sales generated in each region.

Output-

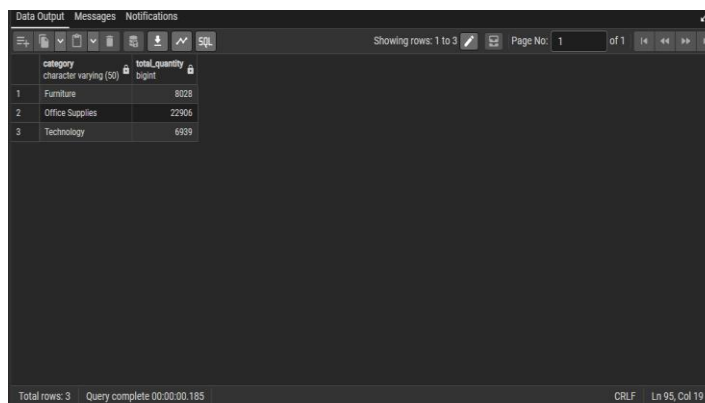


The screenshot shows a SQL query result in a dark-themed interface. The table has two columns: 'region' (character varying (50)) and 'total_sales' (numeric). There are four rows of data. The status bar at the bottom indicates 'Total rows: 4' and 'Query complete 00:00:00.147'.

	region	total_sales
1	West	725457.8245
2	East	678781.2400
3	Central	501239.8908
4	South	391721.9050

2. **Total Quantity by Category-** This query shows the total quantity of products sold in each category.

Output-

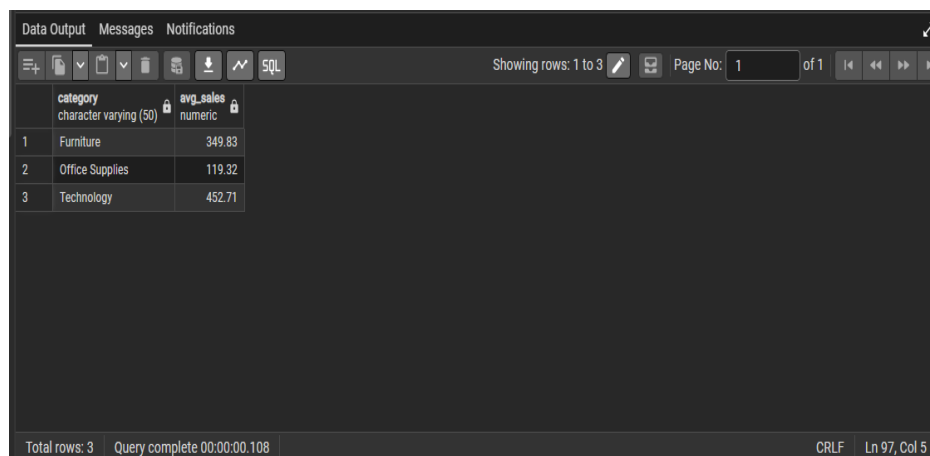


The screenshot shows a SQL query result in a dark-themed interface. The table has two columns: 'category' (character varying (50)) and 'total_quantity' (bigint). There are three rows of data. The status bar at the bottom indicates 'Total rows: 3' and 'Query complete 00:00:00.185'.

	category	total_quantity
1	Furniture	8028
2	Office Supplies	22906
3	Technology	6939

3. **Average Sales by Category-** This query computes the average sales value for each category.

Output-



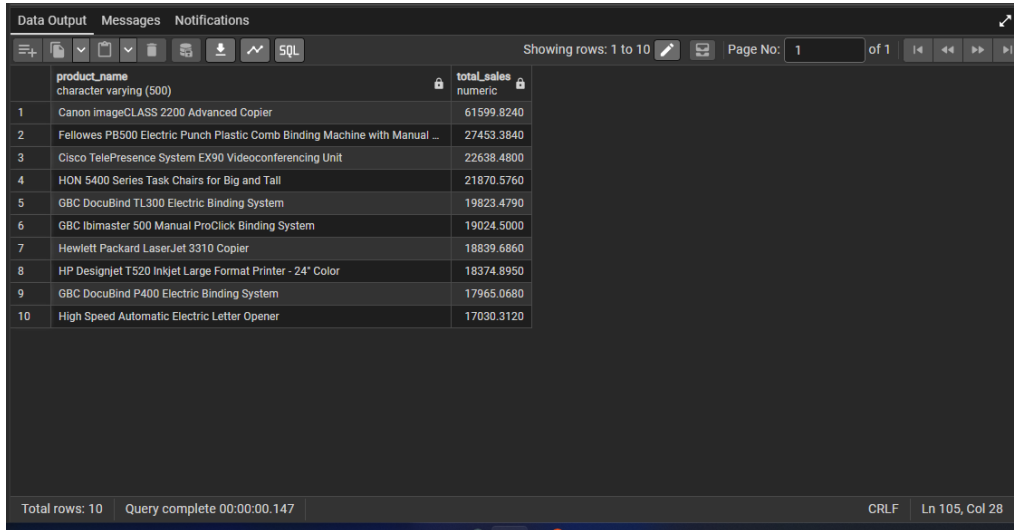
The screenshot shows a SQL query result in a dark-themed interface. The table has two columns: 'category' (character varying (50)) and 'avg_sales' (numeric). There are three rows of data. The status bar at the bottom indicates 'Total rows: 3' and 'Query complete 00:00:00.108'.

	category	avg_sales
1	Furniture	349.83
2	Office Supplies	119.32
3	Technology	452.71

5. Sort and limit results –

1. Top 10 Products by Sales- This query identifies the highest revenue-generating products.

Output-

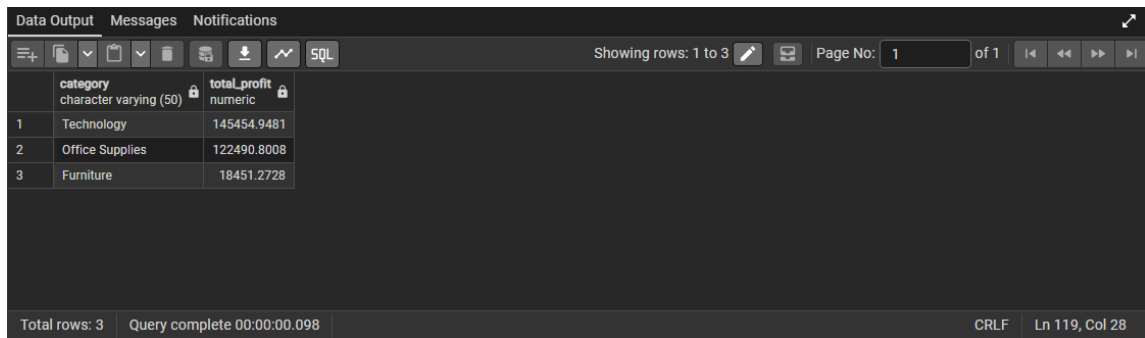


	product_name character varying (500)	total_sales numeric
1	Canon imageCLASS 2200 Advanced Copier	61599.8240
2	Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual ...	27453.3840
3	Cisco TelePresence System EX90 Videoconferencing Unit	22638.4800
4	HON 5400 Series Task Chairs for Big and Tall	21870.5760
5	GBC DocuBind TL300 Electric Binding System	19823.4790
6	GBC Ibimaster 500 Manual ProClick Binding System	19024.5000
7	Hewlett Packard LaserJet 3310 Copier	18839.6860
8	HP DesignJet T520 Inkjet Large Format Printer - 24" Color	18374.8950
9	GBC DocuBind P400 Electric Binding System	17965.0680
10	High Speed Automatic Electric Letter Opener	17030.3120

Total rows: 10 Query complete 00:00:00.147 CRLF Ln 105, Col 28

2. Top Categories by Profit- This query ranks categories based on total profit earned.

Output-



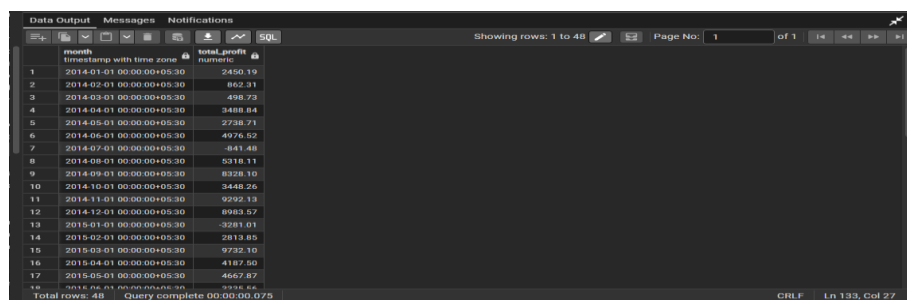
	category character varying (50)	total_profit numeric
1	Technology	145454.9481
2	Office Supplies	122490.8008
3	Furniture	18451.2728

Total rows: 3 Query complete 00:00:00.098 CRLF Ln 119, Col 28

6. Use cases-

1. Monthly Profit Analysis- This query analyzes profit trends month by month.

Output-

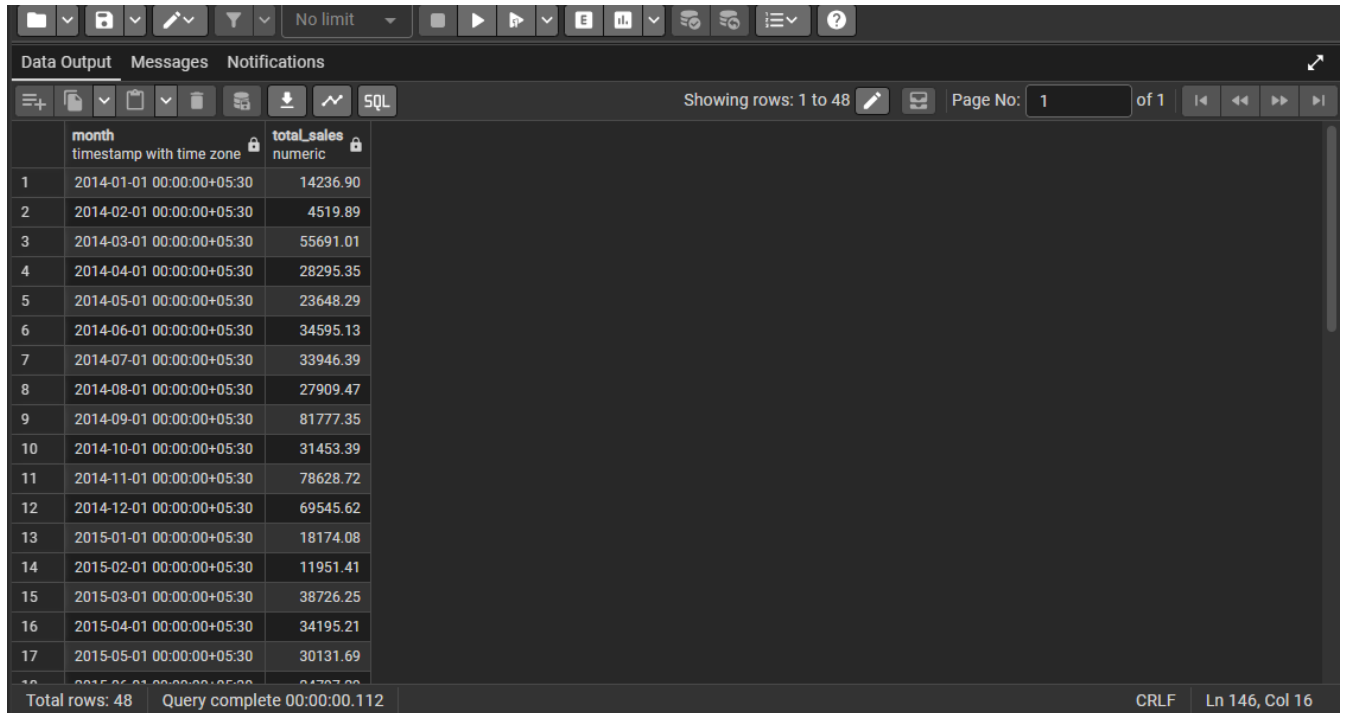


	month timestamp with time zone	total_profit numeric
1	2014-01-01 00:00:00+05:30	2450.19
2	2014-02-01 00:00:00+05:30	862.31
3	2014-03-01 00:00:00+05:30	498.73
4	2014-04-01 00:00:00+05:30	2488.84
5	2014-05-01 00:00:00+05:30	2738.71
6	2014-06-01 00:00:00+05:30	4976.52
7	2014-07-01 00:00:00+05:30	-841.48
8	2014-08-01 00:00:00+05:30	5318.11
9	2014-09-01 00:00:00+05:30	8328.10
10	2014-10-01 00:00:00+05:30	3448.26
11	2014-11-01 00:00:00+05:30	9292.13
12	2014-12-01 00:00:00+05:30	8983.57
13	2015-01-01 00:00:00+05:30	-3281.01
14	2015-02-01 00:00:00+05:30	2813.82
15	2015-03-01 00:00:00+05:30	9732.10
16	2015-04-01 00:00:00+05:30	4187.50
17	2015-05-01 00:00:00+05:30	4667.87
18	2015-06-01 00:00:00+05:30	5118.84

Total rows: 48 Query complete 00:00:00.075 CRLF Ln 133, Col 27

2. **Monthly Sales Analysis**-This query evaluates monthly sales performance over time.

Output-

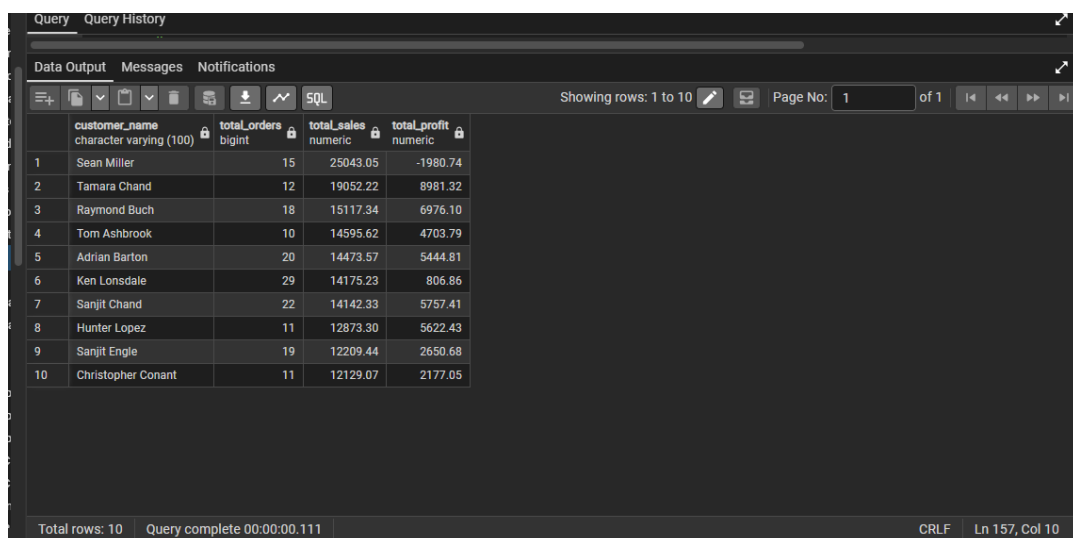


The screenshot shows a database query result in a dark-themed interface. The table has two columns: 'month' (timestamp with time zone) and 'total_sales' (numeric). The data spans from January 2014 to May 2015. The status bar at the bottom indicates 'Total rows: 48' and 'Query complete 00:00:00.112'.

	month timestamp with time zone	total_sales numeric
1	2014-01-01 00:00:00+05:30	14236.90
2	2014-02-01 00:00:00+05:30	4519.89
3	2014-03-01 00:00:00+05:30	55691.01
4	2014-04-01 00:00:00+05:30	28295.35
5	2014-05-01 00:00:00+05:30	23648.29
6	2014-06-01 00:00:00+05:30	34595.13
7	2014-07-01 00:00:00+05:30	33946.39
8	2014-08-01 00:00:00+05:30	27909.47
9	2014-09-01 00:00:00+05:30	81777.35
10	2014-10-01 00:00:00+05:30	31453.39
11	2014-11-01 00:00:00+05:30	78628.72
12	2014-12-01 00:00:00+05:30	69545.62
13	2015-01-01 00:00:00+05:30	18174.08
14	2015-02-01 00:00:00+05:30	11951.41
15	2015-03-01 00:00:00+05:30	38726.25
16	2015-04-01 00:00:00+05:30	34195.21
17	2015-05-01 00:00:00+05:30	30131.69

3. **Top Customers Analysis**- This query identifies customers contributing the highest sales and profit.

Output-

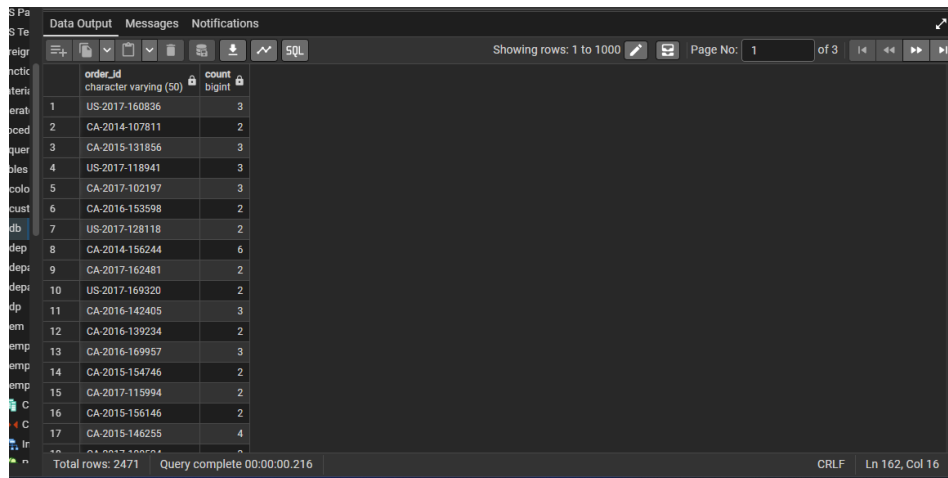


The screenshot shows a database query result in a dark-themed interface. The table has four columns: 'customer_name' (character varying (100)), 'total_orders' (bigint), 'total_sales' (numeric), and 'total_profit' (numeric). The data lists the top 10 customers. The status bar at the bottom indicates 'Total rows: 10' and 'Query complete 00:00:00.111'.

	customer_name character varying (100)	total_orders bigint	total_sales numeric	total_profit numeric
1	Sean Miller	15	25043.05	-1980.74
2	Tamara Chand	12	19052.22	8981.32
3	Raymond Buch	18	15117.34	6976.10
4	Tom Ashbrook	10	14595.62	4703.79
5	Adrian Barton	20	14473.57	5444.81
6	Ken Lonsdale	29	14175.23	806.86
7	Sanjit Chand	22	14142.33	5757.41
8	Hunter Lopez	11	12873.30	5622.43
9	Sanjit Engle	19	12209.44	2650.68
10	Christopher Conant	11	12129.07	2177.05

4. **Duplicate Order Detection**- This query checks for order IDs appearing more than once in the dataset.

Output-



The screenshot shows a SQL query result window with a table containing 17 rows. The table has two columns: 'order_id' (character varying (50)) and 'count' (bigint). The data shows various order IDs and their corresponding counts, ranging from 2 to 6. The status bar at the bottom indicates 'Total rows: 2471' and 'Query complete 00:00:00.216'.

	order_id	count
1	US-2017-160836	3
2	CA-2014-107811	2
3	CA-2015-131856	3
4	US-2017-118941	3
5	CA-2017-102197	3
6	CA-2016-153598	2
7	US-2017-128118	2
8	CA-2014-156244	6
9	CA-2017-162481	2
10	US-2017-169320	2
11	CA-2016-142405	3
12	CA-2016-139234	2
13	CA-2016-169957	3
14	CA-2015-154746	2
15	CA-2017-115994	2
16	CA-2015-156146	2
17	CA-2015-146255	4

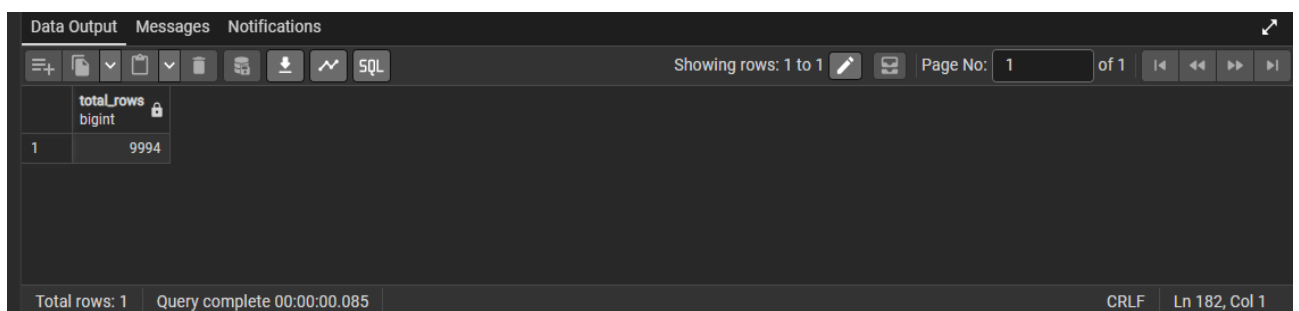
7. **Validate results (Data Quality)-**

1. **Sales Equal to Zero Check**- This query verifies whether any transaction has zero sales value.
2. **Null Value Check**-This query checks for missing values in important columns such as Order ID and Customer Name.

Note: Above 2 queries returned no results because there are no records in the dataset that satisfy the specified condition. The query was executed successfully, but the dataset does not contain matching entries for the given filter criteria.

3. **Row Count Validation**- This query validates the total number of rows present in the dataset after import.

Output-



The screenshot shows a SQL query result window with a single row in a table. The table has one column: 'total_rows' (bigint). The value in the row is 9994. The status bar at the bottom indicates 'Total rows: 1' and 'Query complete 00:00:00.085'.

	total_rows
1	9994